1. Work, Energy and Power

1.1. Introduction

Principle of Conservation of Energy

Energy cannot be created or destroyed; it can only be changed into other forms.

- **Work Done:** $W = Fs$
- **Kinetic Energy:** $E_k = \frac{1}{2}mv^2$
- **Gravitational Potential Energy:** $E_P = mgh$
- **Power:** $P = \frac{W}{T}$ and $P = Fv$

1.2. Changes in Energy

$$\varepsilon_f - \varepsilon_i = (\text{Work})_{engine} - (Work)_{friction}$$

- ε_f is the final kinetic energy of the object
- ε_i is the initial kinetic energy of the object
- $(\text{Work})_{\text{engine}}$ is the energy caused by driving force acting on the object
- $(\text{Work})_{\text{friction}}$ is the energy used up by frictional force or any resistive force

{S05-P04} Question 7:

Car travelling on horizontal straight road, mass 1200kg. Power of car engine is $20kW$ and constant. Resistance to motion of car is $500N$ and constant. Car passes point A with speed $10ms^{-1}$. Car passes point B with speed $25ms^{-1}$. Car takes $30.5s$ to move from A to B.

1. Find acceleration of the car at A
2. Find distance AB by considering work & energy

Solution:

Part (i)

Use formula for power to find the force at A

$$P = Fv$$

$$20000 = 10F \qquad Driving\ force = 2000N$$

We must take into account the resistance to motion

$$\therefore F = Driving\ Force - Resistance = 2000 - 500$$

$$F = 1500$$

Use Newton's Second Law to find acceleration:

$$1500 = 1200a \qquad a = \frac{1500}{1200} = 1.25ms^{-2}$$

Part (ii)

Use power formula to find work done by engine:

$$P = \frac{\text{w.d.}}{t}$$

$$20000 = \frac{\text{w.d.}}{30.5} \qquad w.d. = 610000J$$

There is change in kinetic energy of the car so that means some work done by the engine was due to this:

$$k.E.\ at\ A = \frac{1}{2}1200\,(10)^2 \qquad k.E.\ at\ B = \frac{1}{2}1200\,(25)^2$$

$$Change\ in\ k.E. = \ k.E.\ at\ B - k.E.\ at\ A$$

$$Change\ in\ k.E. = 375000 - 60000 = 315000$$

There is also some work done against resistive force of $500N$; due to law of conservation of energy, this leads us to the main equation:

$$w.d.\ by\ engine = change\ in\ k.E + \ w.d.\ against\ resistance$$

$$610000 = 315000 + 500s$$

$$s = \frac{610000 - 315000}{500} = \frac{295000}{500} = 590m$$

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